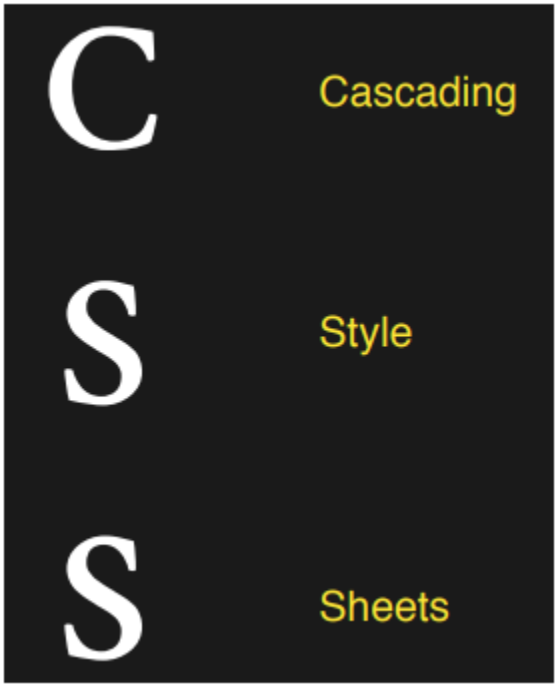
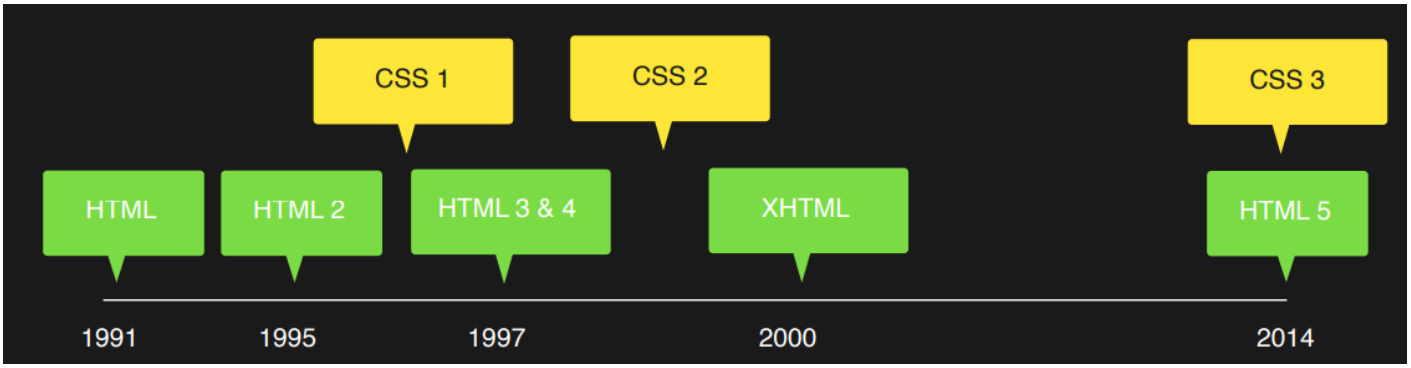
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development





HIRA KUMAR

❖ Units

- CSS units define measurement values used for properties like width, height, margin, padding, font-size, etc.

They are mainly of 2 types:-

1. Absolute Units (Fixed size)
2. Relative Units (Flexible size)

1. Absolute Units (Fixed size)

These units have a fixed value and do NOT change with screen size.

Example –

(a) px = Pixels (most commonly used)
(b) cm = Centimeters
(c) mm = Millimeters
(d) in = Inches (1in = 96px)
(e) pt = Points (1pt = 1/72 inch)
(f) pc = Picas (1pc = 12pt)

✓ Note :- Font-size in html5 is by-default = 16px

2. Relative Units (Flexible size)

These units depend on parent element or screen size

Example –

(a) % = Relative to parent element
(b) em = Relative to parent font-size
(c) rem = Relative to root (html) font-size
(d) vw = 1% of viewport width (browser width)

(e) vh = 1% of viewport height
(f) vmin = Smaller of vw or vh
(g) vmax = Larger of vw or vh
(h) ch = Width of character "0"
(i) ex = Height of letter "x"

Property & Value

- In CSS, properties and value work together as a pair to define the styling of HTML elements. Think of it as a “what” and “how” relationship.

(property कहता हैं “क्या करना हैं” और value कहता हैं “कितना” करना हैं |)

❖ Property

- Properties are predefined keywords in CSS that specify which styling feature you want to modify. Each property accepts specific types of values.

Property categories

Layout	Display, position, width,height
Typography	Font-family,font-size,font-weight
Color & background	Color, background-color
Spacing	Margin, padding
Borders	Border, border-radius
……..	

❖ Value

- Values are the setting you apply to properties. They can be :-

Keywords	Red, center, bold
Numbers with units	16px, 50%, 2em

Color codes	#FF0000, rgb(255,0,0)
URLs	url("image.jpg")
Multiple values	1px solid black
……	

- **Syntax**

```
Selector
    {
        Property:value;
    }
```

- **Example**

```
P
{
    Color:red;
}
```

Here, p= selector, color= property, value= red;

❖ Some property and value

1. Text, Font & Color Property

Color :→ Color properties control the colors of elements on your webpage. CSS offers multiple ways to specify colors.

Text :→ Text properties control the appearance and layout of text content.

Font :→ font property control the appearance font content.

❖ Color property (color:)

Value			
Name value	RGB value (r=red,g=green,b=blue)	Hexadecimal value	RGBA value (r=red,g=green,b= blue,a= alpha)

Red
Blue
Green
Yellow
…….
	Red = (255,0,0)
	Blue=(0,0,255)
	White=(0,0,0)
	Green=(0,255,0)
	……
		Red= #FF0000
		Black=#000000
		White=#FFFFFF
		Green=#008000
		…….
			Alpha means transparency
			Alpha range = 0 to 1
			Where 0= transparent
			1=full color
			(255,0,0,0.5)
			(0,255,0,1)
			……

Example			
color:red;	color:rgb(255,0,0);	Color:#FF0000;	Color:rgba(255,0,0,0.5)

Note :- More Rgb color code / hexadecimal code find, then search **color picker** on browser.

```
  <style>
    .c{
      color: red;
    }
  </style>
  <body>
<!-- color use -->
<p class="c"> My name is hira kumar</p>
<!-- RGB use-->
<a style="color: rgb(0,0,255);"> HTML= Hyper text markup language</a>
<!-- RGBA use-->
<p style="color: rgba(0,255,0,0.3);"> CSS= Cascading style sheet</p>
<!-- hexadecimal use-->
<p style="color: #FF0000;"> JS= Javascript</p>
</body>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

HTML= Hyper text markup language

CSS= Cascading style sheet

JS= Javascript

❖ Font Property

(i) Font-size (font-size:) (Control (Decrease/enlarge) the size of the text)	
Value	**Example**
pixel(px)	"font-size : 50px;"
Percentage(%)	"font-size : 150%;" (a/c to parent font-size)
Element(em)	"font-size:3em;" (let, parent font=5, then 3em=3*5=15)
root element (rem)	"font-size:2rem;"(size fix a/c to root element e.g.- html < style="font-size:10px;"> then 2 rem = 2*10=20)
Keyword Values → Pre-define size	
xx-small, x-small,medium, large,x-large,xx-large	"font-size:large;"

✓ By default- browser font-size is 16px;

Example :-

```html
  <style>
    .c{
      font-size:20px;
    }
  </style>
  <body>
<!-- use pixel -->
<p class="c"> My name is hira kumar</p>
<!-- use percentage (browser use by default-16px)-->
<a style="font-size:50%;"> HTML= Hyper text markup language</a>
<!-- use em-->
<p style="font-size:2em;"> CSS= Cascading style sheet</p>
<!-- use rem-->
<p style="font-size:5rem;"> JS= Javascript</p>
<!-- use keyword-->
<p style="font-size:x-large"> Front-End </p>
</body>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

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CSS= Cascading style sheet

JS= Javascript

Front-End

(ii) Font-family (font-family:) (Control font style text)	
Value	**Example**
Arial	“font-family:arial;”
Verdana	“font-family:verdana;”
Times new roman	“font-family:times new roman;”
Georgia	“font-family: Georgia”
Courier new, trebuchet MS, Tahoma, Impact, Helvetica, Comic Sans MS,….	

✓ By default font-family on browser→ Times New Roman (average case)

❖ What is Font Fallback System

When more than one values used in font-family property, this called font fallback system.

Example :- “font-family:Arial,Georgia,Bell MT,…;”

Example-

```html
<style>
  .c{
    font-family:arial;
  }
</style>
<body>
<!-- use arial  -->
<p class="c"> My name is hira kumar</p>
<!-- use Gill sans font-->
<a style="font-family:Gill Sans;"> HTML= Hyper text markup language</a>
<!-- use fallback system-->
<p style="font-family: Arial, Helvetica, sans-serif;"> CSS= Cascading style sheet</p>
<!-- cursive-->
 <p style="font-family : cursive;"> JS = Java Script</p>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

HTML= Hyper text markup language

CSS= Cascading style sheet

JS = Java Script

(iii) Font-weight (font-weight:) (Control thickness/boldness of text)		
Value	**Example**	**Meaning /use**
Normal (default)	“font-weight:normal;”	Normal
Bold	“font-weight:bold;”	Text को Bold करेगा
lighter	“font-weight-ligher;”	Text को parent से थोड़ा पतला करेगा |
Bolder	“font-weight-bolder;”	Text को parent से थोड़ा मोटा करेगा
Numerica value		
100	“font-weight:100;”	thin

200	“font-weight:200;”	Extra light
300	“font-weight:300;”	Light
400	“font-weight:400;”	Normal
500	“font-weight:500;”	Medium
600	“font-weight:600;”	Semi bold
700	“font-weight:700;”	Bold
800	“font-weight:800;”	Extra bold
900	“font-weight:900;”	black

Example –

```html
<style>
  .c{
    font-weight:bolder;
  }
</style>
<body>
<!-- use bolder  -->
<p class="c"> My name is hira kumar</p>
<a style="font-weight:100;"> HTML= Hyper text markup language</a>
<p style="font-weight:normal"> CSS= Cascading style sheet</p>
 <p style="font-weight:700"> JS = Java Script</p>
 <p style="font-weight:900; color: red;"> Front-End</p>
</body>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

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CSS= Cascading style sheet

JS = Java Script

Front-End

(iv) Font-style (font-style:) (To give italic style to text)		
Value	**Example**	**Meaning /use**
Normal	“font-style:normal;”	Normal

Italic	“font-style:italic;”	For italic
Oblique	“font-style:oblique;”	For italic (slanted tilted)

Example –

```
<style>
  .c{
    font-style:normal;
  }
</style>
<body>
<p class="c"> My name is hira kumar</p>
<a style="font-style:italic;"> HTML= Hyper text markup language</a>
<p style="font-style:oblique; color: red;"> CSS= Cascading style sheet</p>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

HTML= Hyper text markup language

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(v) Font (font:) Shorthand (To write all in one line)
Syntax→ Font: italic bold 18px arial;

Example

```html
<style>
  *{
    font: italic bold 20px arial;
  }
</style>
<body>
<p > My name is hira kumar</p>
<a > HTML= Hyper text markup language</a>
<p > CSS= Cascading style sheet</p>
</body>
```

http://127.0.0.1:5500/index.html

My name is hira kumar

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❖ Line height Property

Adjusts the gap between two line

(vi) Line-height		
Value	**Example**	**Meaning /use**
Normal	“line-height:normal;”	Normal
Number(1.5,3.0,…)	“line-height:2.0;”	Gap 2.0 b/w two lines
Length(px,em,rem)	“line-height:50px;”	Gap 50px
Percentage(%)	“line-height:150%;”	Gap 150% parent line

❖ word-spacing property

- it is use to space b/w two words.
- Syntax –

```
p {
    word-spacing: 10px;
}
```

- Example –

Default

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Incidunt cupiditate earum molestias quam fuga deserunt quos ducimus dicta ratione maxime, aliquid placeat dignissimos ab nihil sint voluptas. Quo, vitae repellat.z

After word-spacing

```
word-spacing:15px;
```

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Incidunt cupiditate earum molestias quam fuga deserunt quos ducimus dicta ratione maxime, aliquid placeat dignissimos ab nihil sint voluptas. Quo, vitae repellat.z

❖ letter-spacing property

- it is use to space b/w letters
- Syntax –

```
p {
    letter-spacing: 2px;
}
```

- Example –

Default

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Incidunt cupiditate earum molestias quam fuga deserunt quos ducimus dicta ratione maxime, aliquid placeat dignissimos ab nihil sint voluptas. Quo, vitae repellat.z

After letter-spacing

```
letter-spacing:5px;
```

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Incidunt cupiditate earum molestias quam fuga deserunt quos ducimus dicta ratione maxime, aliquid placeat dignissimos ab nihil sint voluptas. Quo, vitae